MPPC (S13371-6050CN,-6075CN)

Jul. 30, 2019

SPECIFICATION SHEET

■ Structure

Parameters	Value		Unit
	S13370-6050CN	S13370-6075CN	
Effective photosensitive area	6.0 x 6.0		mm^2
Number of channels	1		-
Pixel pitch	50	75	μm
Number of pixel	14,400	6,400	-
Package	Ceramic		-
Window	Bare (unsealed)		-

■ Absolute Maximum Ratings

Parameters	Symbol	Value	Unit
Operating temperature *1	Topr	up to +60	°C
Storage temperature *1	Tstr	-20 to +80	°C
Soldering condition	Tsol	350°C max., once, 3 sec max.	°C

*1: No condensation

When there is a temperature difference between a product and the surrounding area in high humidity environment, dew condensation may occur on the product surface. Dew condensation on the product may cause deterioration in characteristics and reliability.

■ Electrical and optical characteristics (Typ. T = 25 deg C, Vr = Vop unless otherwise noted)

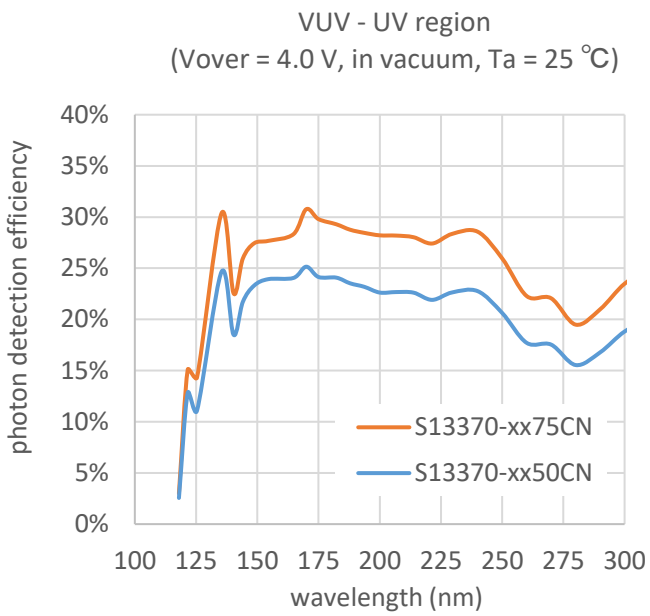
Parameters	Symbol	Value (typ.)		Unit
		S13370-6050CN	S13370-6075CN	
Spectral response range	λ	120 to 900		nm
Peak sensitivity wavelength	λp	500		nm
Photon detection efficiency at λp *2	-	32	40	%
Photon detection efficiency at 175nm (in vacuum) *2	-	24	30	%
Breakdown voltage	V_{br}	53 +/- 5		V
Recommended operating voltage *3	V_{op}	V_{br} + 4.0		V
Dark count rate / channel	DCR	4.0 (MAX:12)		Mcps
Crosstalk probability	P_{CT}	3	5	%
Terminal capacitance at Vop / channel	Ct	1200		pF
Gain	M	2.55×10^6	5.8×10^6	-
Temperature coefficient of reverse voltage Around room temperature	ΔTV_{op}	54		mV/deg C

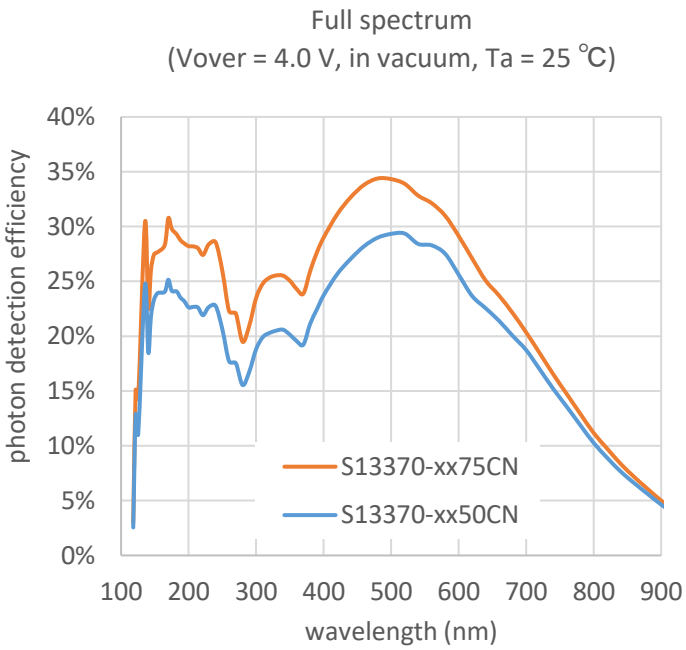
*2 : Photon detection efficiency does not include crosstalk and after pulse.

*3 : Refer to the data attached for each products.

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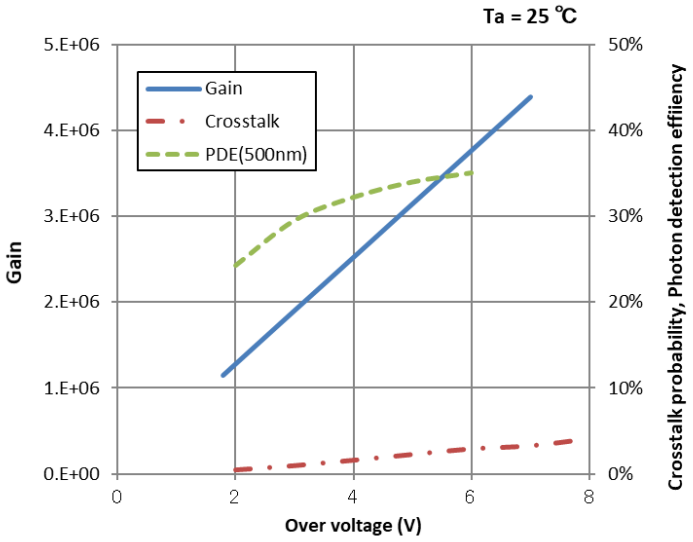
■ Photon detection efficiency v.s. Wavelength (Vr = V_{op} = V_{br} + 4.0V, measurement example)



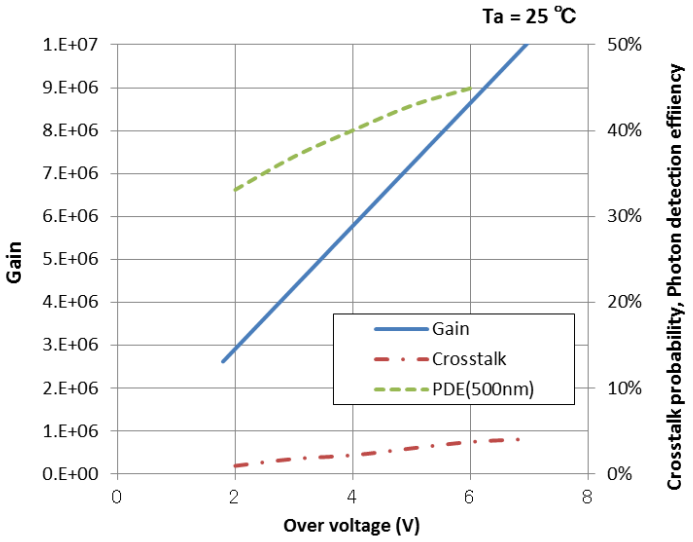


■Overvoltage specifications of gain, crosstalk probability and photon detection efficiency (typical example)

S13370-6050CN

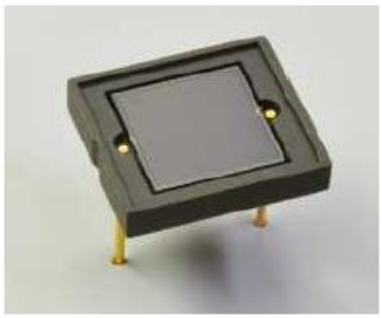


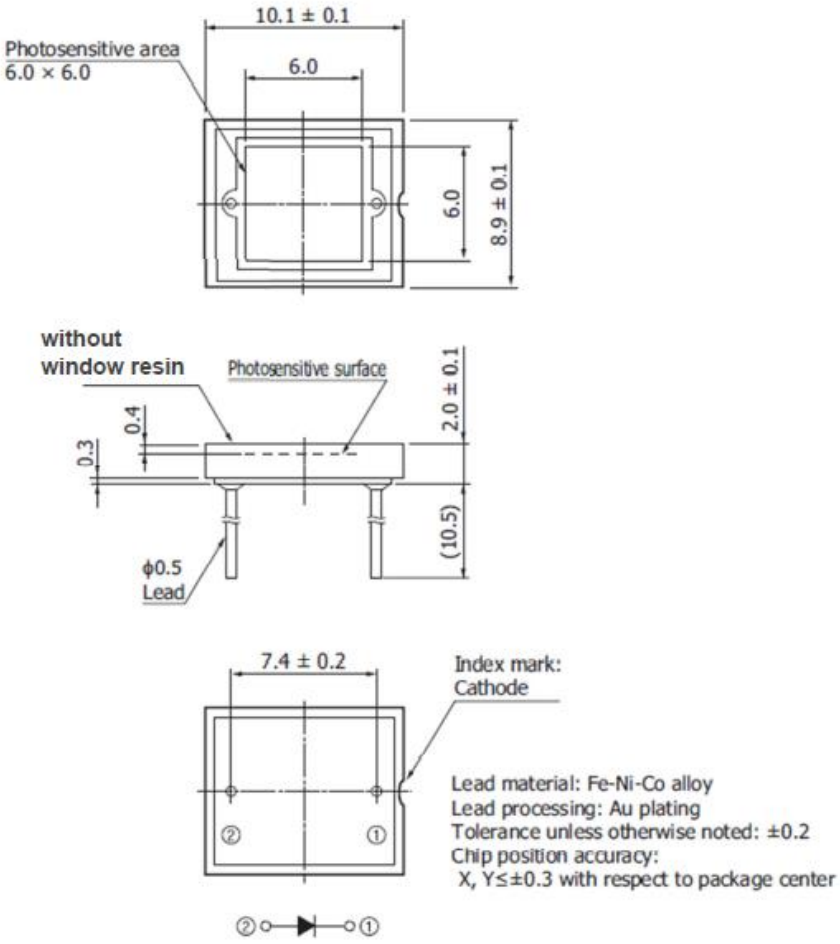
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■ Dimensional outline (unit: mm)





■ Precautions

Please refer below documents for precautions on handling the product.

➢ Precautions for unsealed product

https://www.hamamatsu.com/resources/pdf/ssd/unsealed_KXX-A12014-ENG.pdf